

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 1 – Coding Challenge Task

|                  |                                                                              |               |                                           |
|------------------|------------------------------------------------------------------------------|---------------|-------------------------------------------|
| Course Code:     | CSA06                                                                        | Course Title: | Design and Analysis of Algorithms         |
| CO Assessed:     | CO4: Articulation (BL4, BL5)                                                 | Assessment:   | Assessment Tool 1 – Coding Challenge Task |
| Weightage:       | 40% of CIA                                                                   | Total Marks:  | 100                                       |
| CO4 Statement:   | Solve optimization problems using dynamic programming and greedy strategies. |               |                                           |
| Student Name:    | RUTHIKA REDDY                                                                | Reg. No.:     | 192421138                                 |
| Section / Batch: | BTECH IT                                                                     | Date:         |                                           |

### Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given questions.
- Write clear code for the given problem.
- Analyze the Time complexity and Space complexity of your code using dynamic programming and greedy strategies

| Question Type         | Focus Area                                                                                      | Questions |
|-----------------------|-------------------------------------------------------------------------------------------------|-----------|
| Coding Challenge Task | Focus on Code and analyze optimization problems using dynamic programming and greedy strategies | Q1        |

### SECTION A – Coding Challenge Task

#### Code of Conduct

I \_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: \_\_\_\_\_

### RUBRICS FOR CALCULATION

#### Coding Challenge Task

| Criteria                | Weightage | Level 4 (Exemplary)                                                                       | Level 3 (Proficient)                                      | Level 2 (Developing)                          | Level 1 (Beginning)                              |
|-------------------------|-----------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------|--------------------------------------------------|
| Problem Understanding   | 20%       | Clearly understands problem constraints, edge cases, and competitive requirements         | Understands problem with minor gaps in constraints        | Partial understanding; misses key constraints | Misinterprets problem or constraints             |
| Algorithm               | 25%       | Selects optimal algorithm with best time and space complexity for competitive constraints | Uses efficient algorithm with minor scope for improvement | Uses basic/inefficient approach               | No clear algorithmic approach                    |
| Code Implementation     | 20%       | Writes correct, efficient, and clean code that passes all constraints                     | Code works with minor errors or inefficiencies            | Partially correct code; fails for some cases  | Code does not execute or gives incorrect results |
| Competitive Performance | 20%       | Solves problem within time limits and handles large inputs effectively                    | Solves within acceptable time with minor delays           | Struggles with time limits or larger inputs   | Unable to solve within time constraints          |
| Test Case Handling      | 15%       | Handles all test cases including edge and hidden cases                                    | Handles most test cases                                   | Misses important edge cases                   | Fails on multiple test cases                     |

### Marks Evaluation:

| Criteria                | Wt. | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|-------------------------|-----|---------------------|----------------------|----------------------|---------------------|
| Problem Understanding   | 20% |                     |                      |                      |                     |
| Algorithm               | 25% |                     |                      |                      |                     |
| Code Implementation     | 20% |                     |                      |                      |                     |
| Competitive Performance | 20% |                     |                      |                      |                     |
| Test Case Handling      | 15% |                     |                      |                      |                     |

|                           |  |
|---------------------------|--|
| <b>Total Marks (100):</b> |  |
|---------------------------|--|

| <b>Faculty In-charge</b> | <b>Course Coordinator</b> | <b>Head of Department</b> |
|--------------------------|---------------------------|---------------------------|
| Name:                    | Name:                     | Name:                     |
| Signature: _____         | Signature: _____          | Signature: _____          |
| Date: _____              | Date: _____               | Date: _____               |

**ASSESSMENT CO4-AT1-Coding Challenge Task**[← Back](#)Language: Python C C++ Java**QUESTIONS****Q1**Minimum Cost Path  
100 marks

1/1 answered

**Q1 Minimum Cost Path**

100 marks

A traveler moves through a grid from the top-left corner to the bottom-right corner, and each cell has an associated cost. The traveler may move only right, down, or diagonally down-right. The objective is to find the path with the minimum total cost. Use

**SIMATS**  
ENGINEERING**SIMATS Online Programming Lab****DAA PROGRAMMING****C RUTHIKA REDDY**  
192421138RA

```
2     n = len(grid)
3     m = len(grid[0])
4
5     dp = [[0] * m for _ in range(n)]
6
7     dp[0][0] = grid[0][0]
8     for j in range(1, m):
9         dp[0][j] = dp[0][j - 1] + grid[0][j]
10    for i in range(1, n):
11        dp[i][0] = dp[i - 1][0] + grid[i][0]
12    for i in range(1, n):
13        for j in range(1, m):
```

```
14         dp[i][j] = grid[i][j] + min(  
15             dp[i - 1][j],  
16             dp[i][j - 1],  
17             dp[i - 1][j - 1]  
18         )  
19  
20     return dp[n - 1][m - 1]  
21 if __name__ == "__main__":  
22     import sys  
23     data = sys.stdin.read().strip().split  
24  
25     if not data:  
26         exit()  
27  
28     idx = 0  
29     n = int(data[idx]); idx += 1  
30     m = int(data[idx]); idx += 1  
31  
32     grid = []  
33     for _ in range(n):  
34         row = []  
35         for _ in range(m):  
36             row.append(int(data[idx]))  
37             idx += 1  
38         grid.append(row)  
39  
40     print(min_cost_path(grid))
```

### Input

```
3 3  
1 2 3  
4 5 5  
2 2 0
```

### Output

6

✓ Submitted

© 2026 **SIMATS Online Lab**. All rights reserved.